



Grid connected hybrid renewable energy systems for urban households in Djibouti: An economic evaluation

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ABSTRACT

The cost of electricity produced by thermal power plants in Republic of Djibouti is relatively high at about \$0.32/kWh. This is due to its dependence on imported oil coupled with fluctuating oil prices. Consequently, the customer pays a high electricity bill. However, Djibouti is endowed with indigenous renewable energy resources such as a good solar irradiance of 5.92 kWh/m² day, a potential geothermal energy estimated up to 1000 MW, and few sites with annual wind speed higher than 6 m/s. The goal of this paper is, therefore, to assess an economic evaluation of different grid connected hybrid renewable energy systems to a residential urban house located in Tadjourah city (11.7913° N, 42.8796° E) in the North-Eastern part of Djibouti to reduce the cost of electricity from the grid. To reach this objective, a powerful software tool called HOMER (Hybrid Optimization Model for Electric Renewables) has been used to find the optimum hybrid energy system using real wind and solar irradiation data. The results obtained from this study show that the best economical suited combination of hybrid renewable energy system is a PV-Wind grid connected system. This study shows also that potentially the indigenous renewable energy contribution, in Tadjourah, can be as much as 77 % with 47 % of solar and 30% of Wind energy. The Net Present Cost, the Levelized Cost of Energy, and the operating cost of the optimal HRES are \$337, \$0.002/kWh and \$1,025/year, respectively. When compared with the average cost of grid-only connection of \$0.32/kWh, the optimal hybrid renewable energy system is more economical and will save 51 % of the cost that the customer must pay when using only the electricity from the grid.

1. Introduction

It is well known that most developing countries aiming to eradicate poverty are facing the challenges of finding affordable electricity for their population in both remote and urban areas. It is suggested by a number of authors (Paleta (Paleta et al., 2012), Smith (Smith, 2007) and Pereira (Pereira et al., 2010)) that electricity shortage plays a major role prolonging poverty in the developing countries. By 2030, 89% of Sub-Saharan African population will still have no electricity. 50 % to 75 % of the population of the Republic of Djibouti located in Sub-Saharan Africa region have little or no electricity supply (Walton, 2017). One of the key priorities of the Djibouti government's roadmap is associated with 2030 agenda of sustainable development goals for energy (SDG7) of United Nations (United Nations, 2015). Consequently, the government of Djibouti has placed a stronger emphasis developing a strategic direction and solutions for a cheaper and reliable electricity for its people and this has become one of its keynotes to major financiers and donors.

All electricity produced in Djibouti comes from thermal units running on heavy oil and diesel fuel. The high cost of producing the electricity (USD 0.32/kWh) has become a major burden to the economic and social developments (Guelleh, 2010). However, according to Pillot et al. (Pillot et al., 2015, Pillot et al., 2013b, Pillot et al., 2013a) Djibouti has a substantial daily mean solar irradiation of 5.92 kWh/m². There are also potential sites suitable for a wind energy development such as Gali-Ma'aba, Ghoubbet and Bada Wein with average ground level speeds of 9.1 m/s, 8.9 m/s, and 6.6 m/s, respectively (Awaleh, 2007).

Djibouti Electricity Company (Electricité de Djibouti), (EdD), is a semi-autonomous state-owned company. EdD is responsible for generating, transmitting, distributing, and selling electricity to the customers in Djibouti. So far, the installed electricity generation capacity from EdD is 123 MW with 113 MW from two main power plants called Boulaous and Marabout in Djibouti City. Small provincial power stations are in other districts or regions (Ali-Sabieh, Dikhil, Tadjourah, Obock, and Day) with a total of 10 MW electricity produced. Although EdD has maximum installed capacity of 123 MW, the effective generation capacity is currently limited to 57 MW due to unreliability of old

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Nomenclature

H_{aver}	The monthly average radiation on the horizontal surface of the Earth [kWh/m ² /day]
K_t	The monthly clearness index
$H_{o, ave}$	The extra-terrestrial horizontal radiation [kWh/m ² /day]
$H_{o, ave}$	The monthly average extra-terrestrial radiation
Y_{PV}	The rated capacity of the PV array [kW]
f_{PV}	The PV de-rating factor in %
$\overline{GT}$	The solar radiation incident on the PV array in the current time step [kW/m ²]
$\overline{GT, STC}$	The incident radiation at standard test conditions [1 kW/m ²]
α_p	The temperature coefficient of power [%/°C]
T_c	The PV cell temperature in the current time step [°C]
$T_{c, STC}$	The PV cell temperature under standard test conditions [25 °C]
Z_{hub}	The hub height of the wind turbine [m]
Z_{anem}	The anemometer height [m]
$V_{(Z_{hub})}$	The wind speed at the hub height of the wind turbine [m/s]
$V_{(Z_{anem})}$	The wind speed at anemometer height [m/s]
α	The power law exponent
ρ	The air density ratio can be expressed
Z_m	The site elevation [m]
ρ_o	The air density under standard conditions (The air density at sea level at 1 atm and 60°F is 1.225 kg/m ³)
B	The lapse rate [0.00650 K/m]
Z	The altitude [m]
g	The gravitational acceleration [9.81 m/s ²]
T_0	The standard temperature [288.16 K]
R	Gas constant [287 J/kgK]
$C_{ann, total}$	The total annualized cost of the system [\$/year]
C_{boiler}	The boiler marginal cost [\$/kWh]
$E_{thermal}$	The total thermal load served [kWh/year]
$E_{prim, AC}$	The AC primary load served [kWh/year]
$E_{prim, DC}$	The DC primary load served [kWh/year]
E_{def}	The deferrable load served [kWh/year]

$E_{grid, sales}$	The total grid sale [kWh/year]
C_{tot}	The total annualized cost of the system [\$/Year]
E_{tot}	The total Electricity consumption [kWh/year].
i	The real interest rate in %
n	The number of years
C_{NPC}	The total annual net cost
$CR(i, T_{proj})$	The capital recovery factor
T_{proj}	The project lifetime

Abbreviation

AC	Alternative Current
AfDB	African Development Bank
ANN	Artificial Neural Network
CRF	Capital Recovery Factor
CERD	Centre for Studies and Scientific Research of Djibouti
DC	Direct Current
DJF	Djibouti Franc
EdD	Djibouti Electricity Company (Electricité de Djibouti)
HOMER	Hybrid Optimization Model for Electric Renewables
KW h	Kilowatt-hour
HRES	Hybrid Renewable Energy System
ISERST	Institut supérieur d'Etudes, de Recherches Scientifiques et techniques
kVA	Kilovolt-Ampere
LCOE	Levelized Cost of Electricity
NREL	National Renewable Energy Laboratory
NPC	Net Present Cost
O&M	Operation and Maintenance
PV	Photovoltaic
SP	Subscribed power
SDG	Sustainable Development Goal
VS	Versus
USD	United States Dollar
Yr	Year
W h	Watt-hour
WT	Wind Turbine

generators. Besides, due to the high electricity cost, illegal connection to the grid has become more routine. This problem of theft and non-payment of electricity bills represent 16 % of electricity produced. Moreover, high transmission and distribution losses, poor revenue collection from big companies, government departments, and government owned companies add further strains and challenges to EdD (IRENA, 2015). Despite of that, Djibouti is aiming to generate 100% of its electricity from renewable energy by 2030 (Idriss et al., 2019).

Most studies that have been presented in the literature, particularly in Africa Sub-Saharan countries, were mainly focused on stand-alone HRES (Bekele and Tadesse, 2012, Himri et al., 2008, Nfah and Ngundam, 2009, Kenfack et al., 2009, Kusakana, 2014, Kusakana and Munda, 2009, González et al. (2015) observed that research on HRES connected to the grid has been scarce in literature. Moreover, each study has limited purpose of optimization related to HRES connected to the grid. For example, Mohammad et al. (2017) minimized the life cycle cost (LCC) of the HRES and to maximise the system probability measured in terms loss of power supply probability (LPSP). Alharthi et al. (2018) have performed the technical and economic analyses of the grid-connected Solar PV-Wind Hybrid System of four different cities in the Kingdom of Saudi Arabia using an average community load demand to choose the city with the lowest net present cost (NPC) and levelized the cost of energy (LCOE). Akbari et al. (2019) investigated the techno-economic assessment of a grid-connected hybrid of solar-wind

energy system for a residential urban area in a town located in Iran. Even though the limited space in the residential urban area was not taken into consideration, they ended up choosing an optimized model for a community-based optimal hybrid system consisting of four wind turbines connected to the grid.

Although there has been already a limited published research to date investigating the techno-economic evaluation of grid-connected HRES, few researchers have done grid-connected HRES researches on residential urban households such as Li (2019), Barakat et al. (2016a), Maleki et al. (2017), Barakat et al. (2016b), Liu et al. (2012), Barakat et al. (2022), Gallardo et al. (2020), Samy et al. (2020), Tarigan et al. (2015). Previous research on HRES of grid-tied urban households has not comprehensively been considered to reduce further the cost of grid electricity. This paper extends, therefore, the previous research done by assessing the potential reduction in the overall cost of the installation in terms of economic consideration and optimizing the proposed HRES, and hence this will offer the hybrid renewable energy solutions in Republic of Djibouti. Besides, general research in PV-Wind power generation system of urban households connected to the grid is still limited in developing countries. Such research has been done particularly by Gallardo et al. (2020), and Elnozahy and Abdel-Salam (2019). In this regard, this paper particularly focuses on how these urban households connected to the grid can benefit the combination of these PV-Wind generators to reduce the shortage of grid electricity and satisfy

reliably the load of the households. Consequently, this research will complete and level up this gap. Furthermore, this study will perform with real load demand profile taken straight from real customer who lives in the town where the study is carried out. Real data set will be used despite that some accessed studies in literature do not include real time data according to [González et al. \(2015\)](#), and consequently this may weakens the techno-economic analysis due to the lack of accuracy.

In the context of Republic of Djibouti, the objective of this study is to reduce the amount of electricity purchased from EdD power grid by evaluating the economic feasibility of different hybrid renewable energy systems for an urban household located in Tadjourah of 45 000 habitants. During the summer and peak hours, the EdD distribution grid, which is an isolated power system of 6 diesel generators (600kVA) a total generating capacity of 2.2 MW for Tadjourah city, cannot meet the load demand of the entire city ([RCREEE, 2016](#)). Thus, the electricity generated from hybrid renewable energy system will supply the load demand for the household first and the excess electricity will be injected to the grid. Besides, as far as the authors are aware, no study has ever been presented so far in Djibouti by choosing a suitable model of hybrid renewable energy system optimization for a grid-tied urban household to reduce the electricity bill and optimize the availability of electricity.

The paper is structured as follows: [Section 2](#) provides an overview of hybrid renewable energy systems. [Section 3](#) discusses different methodologies for optimization of HRES and describes HOMER software. [Section 4](#) describes the system configuration of this study. [Section 5](#) evaluates solar and wind resources in Djibouti. [Section 6](#) discusses the output parameters, technical specification of solar PV modules, and wind turbines for this study. [Section 7](#) gives the results and discussions, and conclusion and recommendations are given in [Section 8](#).

2. Hybrid Renewable Energy Systems: An overview

Solar and wind energies are intermittent renewable energy sources except that geothermal energy which is constant and stable but expensive. PV and wind energies alone cannot supply on 24 hours as a base-load, because the sun does not shine 24 hours and the wind does not always blow away. To overcome the weakness of stand-alone energy system such as solar or wind energy, a combination of one or more energy resources is required to make a hybrid renewable energy system (HRES) ([Bajpai and Dash, 2012](#), [Hosseinalizadeh et al., 2016](#)). Combining solar energy and wind power, or batteries with the grid power can form a more cost-effective HRES to produce a reliable power which is available for 24 hours a day. This form of HRES will not be an optimal solution without the right combination of energy resources.

Many studies on HRES have been reported combining different renewable energy resources ([Ashok, 2007](#), [Erdinc and Uzunoglu, 2012](#), [González et al., 2015](#), [Himri et al., 2008](#), [Kamel, 2005](#), [Khare et al., 2016](#), [Lau et al., 2010](#), [Nandi and Ghosh, 2009](#), [Rehman and Al-Hadhrani, 2010](#), [Rehman et al., 2007](#), [Shaahid and Elhadidy, 2007](#), [Shezan et al., 2016](#), [Sedighi et al., 2017](#)). These studies investigated different configurations of hybrid energy systems such as Hydro-PV-Diesel, PV-Wind-Diesel, Hydro-PV-Wind, or PV-Wind/Hydrogen/Fuel cell-based systems. They all had one common goal: increasing efficiency of electricity production, and particularly its reliability, and its availability in both remote and urban areas.

HRES are now closing the gap for electricity shortage in remote areas. A study combining solar wind and diesel generator has been performed by Adaramola et al. ([Adaramola et al., 2014](#)) in order to investigate the techno-economic feasibility for a power supply in remote area of southern Ghana. As a result, using Levelized Cost of Electricity (LCOE) and Net Present Cost of the system (NPC), six economically feasible configurations of hybrid energy systems were identified giving the opportunity for policy makers and all stakeholders in Ghana to shape their strategic energy goal.

[Kenfack et al. \(2009\)](#) have elaborated a method of optimization for a small size of hydro-PV-hybrid system to solve rural electrification

Table 1

Different methodologies for optimization and sizing of HRES.

Authors	Methodologies	Brief description of study
del Real et al. (2009)	Hub concept or energy hub	Optimization strategy for sizing hybrid power systems
Habib et al. (1998)	Design space-based approach	Sizing of solar/wind hybrid system
Margeta and Glasnovic (2010)	Dynamic programming model	Optimal sizing of PV power plant
Dufo-López and Bernal-Aguistín (2008)	Evolutionary Algorithm	to find the best combination of components of complex PV–wind–diesel–hydrogen–battery system
Diaf et al. (2008)	Iterative and probabilistic approaches	Sizing and techno-economical optimization of a stand-alone hybrid photovoltaic/wind system (HPWS) with battery storage
Garcia and Weisser (2006)	Linear programming methodology	determining the size of grid units and dispatch in a wind-diesel power system with hydrogen storage.
James D. Clark (2009)	Matrix approach	sizing of components in distributed renewable energy systems containing multiple sources of power
Mellit et al. (2005)	Adoptive Artificial Neural Network (ANN)	To find a suitable model for sizing Stand-Alone Photovoltaic (SAPV) systems
Kaldellis et al. (2006)	Parametric and numerical approaches	to estimate the optimum dimensions of a hybrid wind-diesel stand-alone system based on the minimum long-term electricity production cost
Ashok (2007)	Quasi-Newton Algorithm	to find an optimal combination of energy components for a typical rural community minimizing the life cycle cost
Ekren and Ekren (2008)	Response surface methodology (RSM)	Sizing optimization of an autonomous PV/wind integrated hybrid energy system with battery storage
Lagorse et al. (2009)	Genetic algorithms then simplex algorithms	to evaluate the validity of the different hybrid configurations.
Tan et al. (2010)	Stochastic method	Sizing of photovoltaic generators and develop a methodology for sizing of stand-alone photovoltaic systems
Samy et al. (2019)	A flower pollination optimization algorithm	Estimating the optimum number of both PV panels and the FC/electrolyzer/H2 storage tanks

problems in a village in Cameroon. They combined a micro-hydro and solar PV as sources of energy as most of hybrid systems proposed in the past were based either on PV-diesel system or PV-Wind systems. They found that implementing a hybrid energy system in remote village is an ideal tool to validate the great renewable potential in Cameroon. They concluded that these results could be used in Cameroon's National Energy Action plan for the provision of energy services in the key sectors involving the fight against poverty.

[Shaahid and El-Amin \(2009\)](#) performed a technique to assess a techno-economic feasibility of hybrid PV-Diesel-Battery power system for a typical Village in Saudi Arabia. As a result, they found that the percentage fuel saving by using hybrid PV-diesel-battery systems is 27% as compared to a diesel only situation. They showed that even though Saudi Arabia is blessed with fossil fuels, the potential of renewable energy option decreasing thousands tons/year of carbon emissions should not be overlooked.

In term of energy management in grid-connected hybrid renewable energy systems, [Olatomiwa et al. \(2016\)](#) found that most studies of hybrid systems were only standalone hybrid systems and recommended that further studies between the sources of hybrid energy systems and the load connected to the grid were essential.

Managing different hybrid renewable energy systems (HRES) with multiple sources of renewable generators is complex. It requires

Table 2

Daily power demand for a middle-class household at Tadjourah (author's own data).

Household appliances	Number of items	Total of Hourly Consumption in Watts	Total electricity demand per day (Wh/day)
Fluorescent tube (40 W)	10	400	2640
Ceiling fan (80 W)	5	400	3920
TV Plasma (120W)	1	120	1320
Split air conditioner (2400 W)	2	4800	26400
Blender (Moulinex) (500 W)	1	500	125
Microwave (700 W)	1	700	175
Desktop PC (90 W)	1	90	225
Fridge (250 W)	1	250	6000
Washing machine (500 W)	1	500	1250
Iron (1100 W)	1	1100	825
Mobiles charging (18 W)	-	18	126
Total	24	8878	43006

powerful tools for their optimization.

3. Optimization of hybrid renewable energy systems

3.1. Different methodologies for optimization of HRES

The objective of the optimization is to find out the most economical planning of the right combinations of hybrid renewable energy systems to satisfy the load demand. To find the optimal combination of stand-alone or grid-connected hybrid energy systems in terms of sizing, various studies in literature review have used different methodologies or procedures as shown in Table 1. In terms of software technicality, (Sinha and Chandel, 2014) have reviewed 19 software: HOMER, Hybrid2, RETScreen, iHOGA, INSEL, TRNSYS, iGRHYSO, HYBRIDS, RAPSIM, SOMES, SOLSTOR, HySim, HybSim, IPSYS, HySys, Dymola/Modelica, ARES, SOLSIM, and HYBRID DESIGNER. They found that the most widely used software tool for such HRES studies is HOMER (Hybrid Optimization Model for Electric Renewables).

3.2. HOMER software

In this study, HOMER is used for the design of different hybrid Renewable energy systems. HOMER has been designed by the United States National Renewable Energy Laboratory (NREL). It is a general-purpose hybrid system software that is very useful for designing a decentralized electric power system. The software designs an optimal power system to fulfil the requirements of the desired loads. HOMER has been used extensively in literature for hybrid renewable energy systems in various studies including (Al Garni et al., 2018, Aziz et al., 2019, Bahramara et al., 2016, Çetinbaş et al., 2019, Kusakana and Munda, 2009, Kord et al., 2010, Shezan et al., 2016, Kusakana, 2014, Guelleh et al., 2020). HOMER uses an economic metric analysis to optimize hybrid or standalone system and ranks the systems according to their net present cost (NPC), Levelized Cost of Energy (LCOE), and operating cost. However, various studies have performed different methodologies optimizing hybrid renewable energy systems. Table 1 summarises different optimization techniques for sizing HRES.

Table 3

Hourly load demand for a typical middle class household in Djibouti (author's own data).

Hour	Load (kW)	Hour	Load (kW)
00:00 – 01:00	2.984	12:00 - 13:00	3.386
00:00 - 02:00	2.984	13:00 - 14:00	5.652
02:00 - 03:00	2.984	14:00 - 15:00	3.050
03:00 - 04:00	2.984	15:00 - 16:00	0.877
04:00 - 05:00	2.984	16:00 - 17:00	0.628
05:00 - 06:00	0.584	17:00 - 18:00	1.200
06:00 - 07:00	0.788	18:00 - 19:00	0.947
07:00 - 08:00	0.925	19:00 - 20:00	3.210
08:00 - 09:00	1.110	20:00 - 21:00	1.060
09:00 - 10:00	1.118	21:00 - 22:00	1.154
10:00 - 11:00	0.918	22:00 - 23:00	1.373
11:00 - 12:00	0.990	23:00 - 00:00	3.136

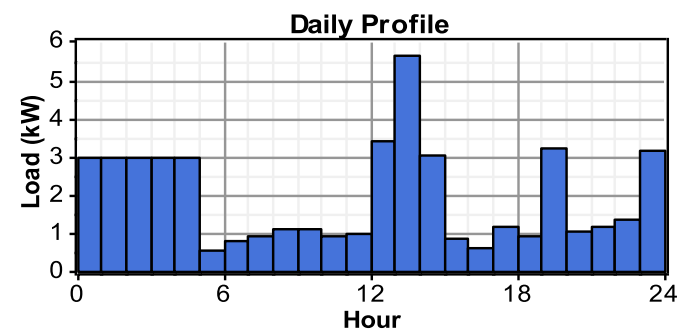


Fig. 1. Average daily load profile for a middle class household in Tadjourah.

4. System configuration

4.1. Site location

This study was carried out to look at one particular dwelling in the city of Tadjourah which lies in the north of the Republic of Djibouti on a latitude of 11.7913° North and longitude of 42.8796° East at an elevation of 4 meters above sea level. Lying on the gulf of Tadjourah, the city is home for 45 000 habitants. The dwelling in question is a middle-class household which has 3 bedrooms, one large dining and living room. It has a concrete roof where a suitable hybrid renewable energy system can be installed.

4.2. Electric load profile for an average middle-class household

An electricity load profile is an important step for the whole study under investigation. The Table 2 gives an estimation of subscribed power for a middle class household in Tadjourah city in Djibouti. The subscribed power (SP) or the real power performing the work used by the appliances' household is estimated at 8878 W. This is the electrical power needed if all electrical appliances are used at once and it is measured in Watts.

In a hot and semi-arid country like Djibouti, the most challenging energy consumption is the air conditioning. A daily load profile of a typical middle-class residential household is estimated to be a scaled average of 43 kWh/day with a scaled peak of 9.2 kW (Table 2). The high demand occurs during mid-day and in the evening for cooling purpose as illustrated in Table 3 and Fig. 1.

With HOMER generating randomly a typical load profile, one can specify different daily profile for each month or just specify a single profile for the whole year. For example, in Djibouti, one can use a different load profile from April to August which are the hottest months of the year. In this study, an assumption has been made and a single load profile has been chosen for the whole year to make it simple as illustrated in Fig. 2.

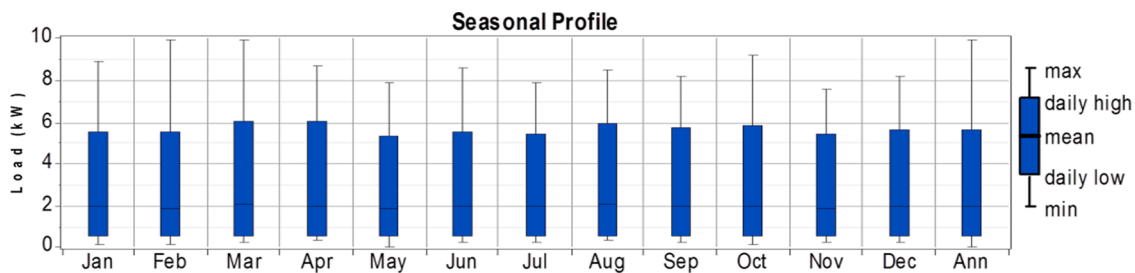


Fig. 2. Seasonal load profile of an urban household in Tadjourah City.

Table 4

Electricity Tariff from 1st March 2016 in Djibouti (author's own data).

Social Category 1	Apparent Power (S)	Price per 1 st Category		Price per 2 nd Category		Range of thickness per Category	
	S = 1 kVA	\$/kWh	DJF/kWh	\$/kWh	DJF/kWh	1 st Cat. (kWh)	2 nd Cat. (kWh)
Social Category 2	S = 3 kVA	0.23	40	0.31	55	200	200+
	S = 6 kVA						
Domestic	S = 9 kVA	0.27	48	0.31	55	210	200+
Domestic	S ≥ 9 kVA	0.27	48	0.31	55	105+ (5*SP)	Extra

Table 5

Typical monthly electricity bill for a middle-class household in Djibouti (author's own calculation).

Account	Quantity	Price per Unit (DJF)	Amount (DJF)
Standby charge Domicile 9 (x days)	30	35.63	1063.9
Tranche 1: Domicile 9 (x kWh)	210	48	10080
Tranche 2: Domicile 9 (x kWh)	1094.51	55	60198.05
Total amount before VAT: 71341.95 DJF VAT Waived: 0			
Total net to pay: 71241.95 DJF; (UD\$ 1 = 177.5 DJF); Total = \$401.36			

4.3. Monthly middle-class household bill in Djibouti

To understand better, in this study, whether the customer is going to save some money, the electricity purchased from EdD must be known to compare later how much the grid connected hybrid energy system could save for the customer. Assumptions have been made that the customer consumes the same daily electrical load for every day for the whole year. Every month this household is assumed to consume 1304.51 kWh per month ((43 kWh/day × 7 days /week × 52 weeks): 12 months). Therefore, taking into consideration of the electricity tariff from EdD illustrated in Table 4, the following Table 5 gives the result of annual bill to be paid to the electricity producer, EdD.

The total monthly bill comes to \$401.36. The annual power consumption of this typical urban household would be 15,654.184 kWh/year. The customer pays \$4,816.32/year to the energy provider, EdD. To lower this energy bill, hybrid renewable energy systems may be the only recourse.

In the next section, the authors are going to evaluate the renewable energy resources in Djibouti such as solar and wind energy.

5. Solar and wind resources in Djibouti

5.1. Solar resources in Djibouti

This study uses solar irradiance data obtained from Pillot et al., (2015) who, developing a satellite-based solar atlas with the ground measurements, demonstrated that Djibouti has a great potential of solar resources, and regarded as one the most important potential in the world with a daily mean irradiation of 5.92 kWh/ m² day as illustrated in Figs. 3 and 4. Fig. 3 represents the yearly map of the daily irradiation

(kWh/m².day) over the Republic of Djibouti, and Fig. 4 represents the average energy per square meter of the daily irradiation that is available from solar source in different months. As one can observe from the Fig. 4, the solar radiation from March to June is much stronger than that of other months followed by the months July to October. The reason is explained in Table 6 and Fig. 5 showing the monthly solar radiation profile and clearness index. As it can be noticed in Fig. 5, the clear index drops after the month of June because of wind blowing from West to East bringing dust called "Kamsin" in Djibouti in this period. The clearness index has a high value under clear sky, sunny conditions, and low value under cloudy conditions. The clearness can be defined on an hourly or monthly basis as shown in Fig. 5 further below.

The clearness index values in HOMER's solar resource inputs window are monthly average values. It is expressed in the following equation with the symbol for the monthly clearness index, K_t .

$$K_t = \frac{H_{ave}}{H_{o,ave}} \quad (1)$$

Where:

H_{ave} is the monthly average radiation on the horizontal surface of the Earth [kWh/m²/day], $H_{o,ave}$ is the extra-terrestrial horizontal radiation, meaning the radiation on the horizontal surface at the top of the earth's atmosphere [kWh/m²/day]. $H_{o,ave}$ is the monthly average extra-terrestrial radiation. So, if either H_{ave} or K_t , are known, then other will be calculated using the above equation.

However, one can agree with (Lau et al., 2010) highlighting that solar energy will only be produced during the daytime; from 6 a.m. to 6 p.m. at Tadjourah site in this study, and at night, electricity will have to be produced either from Wind turbine, battery bank or supplied from the grid.

5.2. Wind resources in Djibouti

A wind resource monitoring project was done through the five districts of Djibouti, in 1983-84 (Awaleh, 2007). The value of monthly mean wind speed at 10 meters at an anemometer height above the ground presented in this study in Fig. 6 was provided by ISERST (Institut supérieur d'Etudes, de Recherches Scientifiques et techniques) replaced by Centre for Studies and Scientific Research of Djibouti (Centre des Etudes et de Recherche de Djibouti), (CERD). The Table 7 indicates the monthly mean wind speeds (m/s) at 10 meters above the ground in Tadjourah.

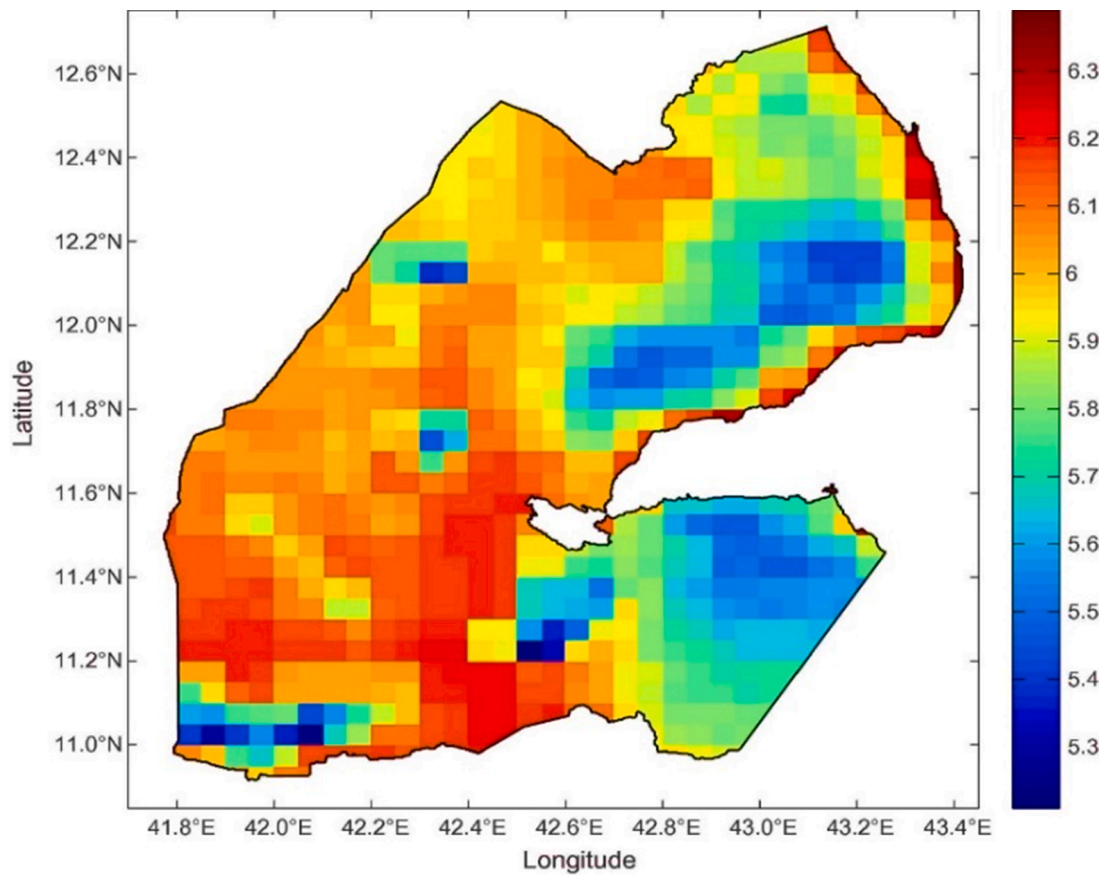


Fig. 3. Yearly map of the daily mean irradiation (kW h/m²/day) over Djibouti (Pillot et al., 2015, Pillot et al., 2013b, Pillot et al., 2013a).

By using HOMER software, the baseline is set of 8760 values representing the average wind speed, expressed in meters per second, for each hour of the year. Then, HOMER displays in Fig. 6 the monthly average wind speed of the site calculated from the baseline data in the wind resource from Table 7 provided by ISERST in 1983-84 (Awaleh, 2007).

In reference with (Awaleh, 2007), Tadjourah district has registered a higher annual mean wind speed of 7.1 m/s comparing to other districts, i.e. Djibouti-ville (3.5 m/s), Ali-Sabieh (3.8 m/s), Dikhil (4.6 m/s), Obock (3.7 m/s). Fig. 7 highlights this difference by using monthly mean wind speeds.

6. Output parameters and components specification

6.1. Solar PV outputs' parameters

The following equation will be used by HOMER to calculate the output of the Solar PV (Adaramola et al., 2014, Hiendro et al., 2013, Rohani and Nour, 2014):

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\overline{G}_T}{\overline{G}_{T, STC}} \right) [1 + \alpha_p (T_C - T_{C, STC})] \quad (2)$$

Where:

Y_{PV} is the rated capacity of the PV array, meaning its power output under standard test conditions [kW], f_{PV} is the PV de-rating factor in %, $\overline{G}_T$ is the solar radiation incident on the PV array in the current time step [kW/m²], $\overline{G}_{T, STC}$ is the incident radiation at standard test conditions [1 kW/m²], α_p is the temperature coefficient of power [%/°C], T_C is the PV cell temperature in the current time step [°C], and $T_{C, STC}$ is the PV cell temperature under standard test conditions [25 °C].

For this study, we assumed that there is no effect of temperature on the PV array. Therefore, HOMER will assume that the temperature co-

efficient of power is zero, so that the above equation simplifies to:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\overline{G}_T}{\overline{G}_{T, STC}} \right) \quad (3)$$

The standard test conditions (STC) mean a radiation of 1 kW/m², a cell temperature of 25°C, and no wind. Thus, the annual energy which is a scaling factor that HOMER applies to the PV array power output to account (such as soiling of the panels, wiring losses, shading, and aging etc..) for reduced output in real world operating conditions compared to the conditions under which the PV panel is rated.

6.2. Wind turbine outputs' parameters

To calculate the output power of the wind turbine at a particular hour, HOMER follows three step processes:

a) It takes the hour's wind speed from the wind resource and adjusts it to the hub height using either two of the following equations:

i) logarithmic profile equation (or log law)

The logarithmic equation assumes that the wind speed is proportional to the logarithm of the height above the ground. The following equation therefore gives the ratio of the wind speed at hub height to the wind speed at anemometer height as highlighted in (Masters, 2004, Ullah et al., 2010):

$$\frac{V(Z_{hub})}{V(Z_{anem})} = \frac{\ln(Z_{hub}/Z_0)}{\ln(Z_{anem}/Z_0)} \quad (4)$$

Where:

Z_{hub} is the hub height of the wind turbine [m], Z_{anem} is the anemometer height [m], Z_0 is the surface roughness length [m], $V(Z_{hub})$ is

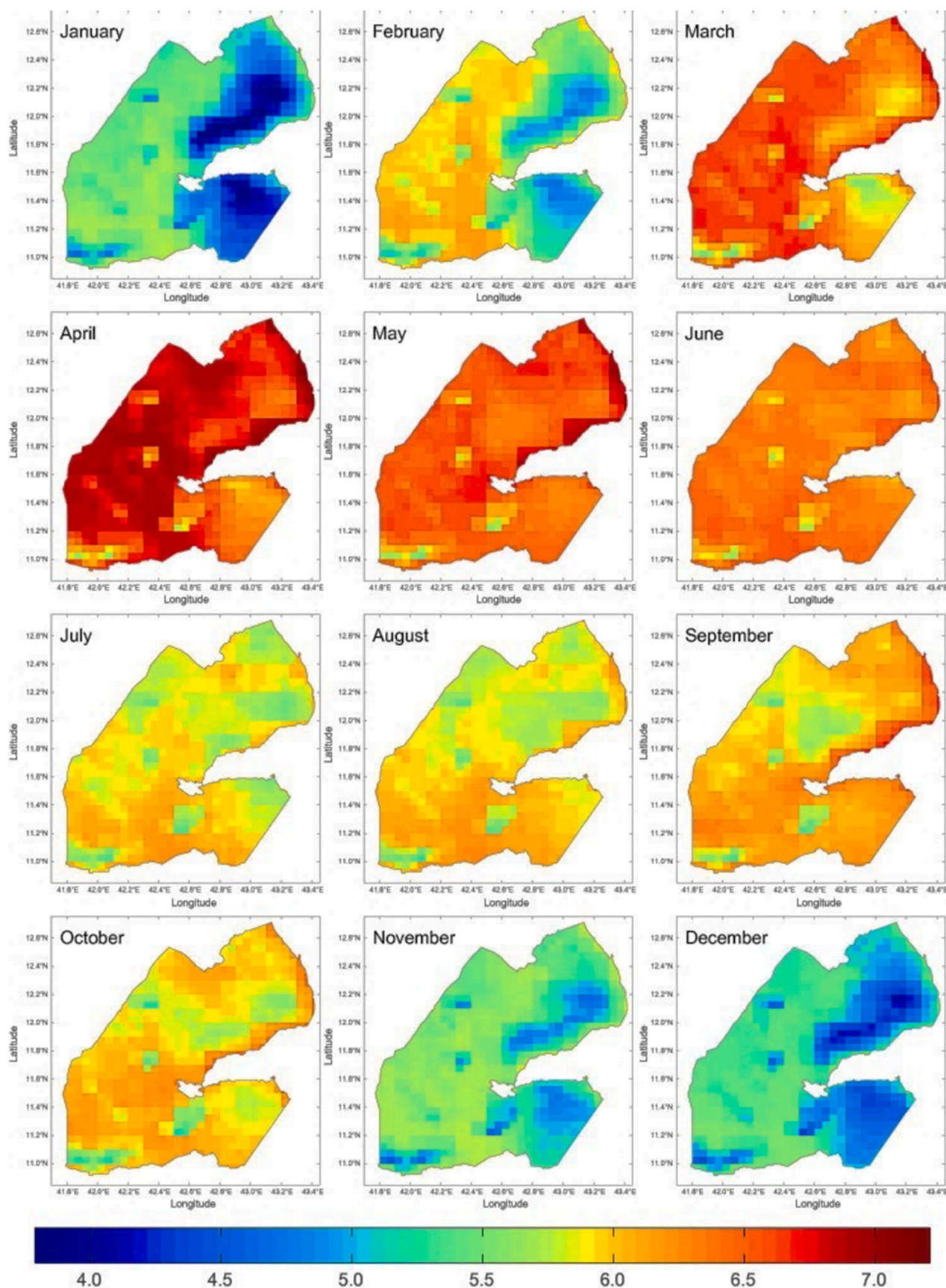


Fig. 4. Monthly mean maps of the daily irradiation (kW h/ m² day) over Djibouti (Pillot et al., 2015).

Table 6

Monthly average daily global irradiation with clear index in Djibouti (Pillot et al., 2015).

Month	Clearness Index	Daily Radiation (kWh/m ² /d)	Month	Clearness Index	Daily radiation (kWh/m ² /d)
January	0.587	5.080	July	0.565	5.890
February	0.603	5.660	August	0.565	5.920
March	0.631	6.400	September	0.594	6.070
April	0.635	6.690	October	0.623	5.960
May	0.617	6.500	November	0.614	5.400
June	0.609	6.340	December	0.607	5.090

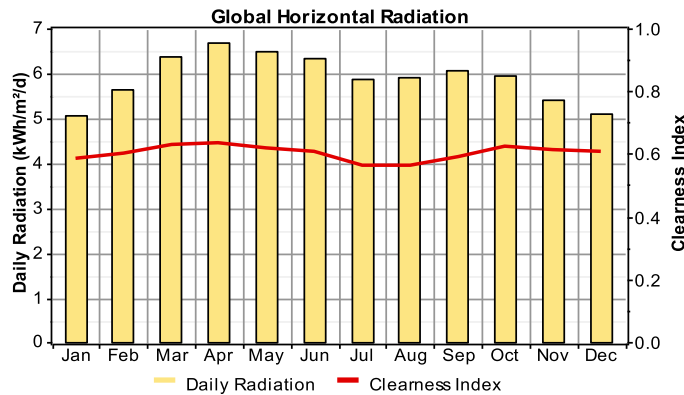


Fig. 5. Global solar radiation of Djibouti, clearness index. Data from (Pillot et al., 2015, Pillot et al., 2013b, Pillot et al., 2013a).

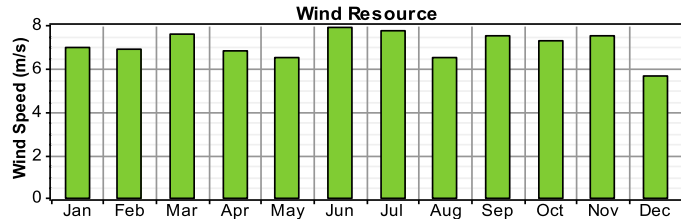


Fig. 6. Monthly average wind speed at Tadjourah district. Data from (Awaleh, 2007).

Table 7

Monthly mean wind speeds (m/s) in Tadjourah city, Djibouti. Data from (Awaleh, 2007).

Month	Wind Speed (m/s)	Month	Wind speed (m/s)
January	7.00	July	7.800
February	6.90	August	6.500
March	7.600	September	7.500
April	6.800	October	7.300
May	6.500	November	7.500
June	7.90	December	5.700

the wind speed at the hub height of the wind turbine [m/s], $V_{(Z_{anem})}$ is the wind speed at anemometer height [m/s], ii) Power law profile equation

The power law profile assumes that the ratio of wind speeds at different heights is given by the following equation (Masters, 2004):

$$\frac{V(Z_{hub})}{V(Z_{anem})} = \left(\frac{Z_{hub}}{Z_{anem}} \right)^{\alpha} \quad (5)$$

Where:

Z_{hub} is the hub height of the wind turbine [m], Z_{anem} is the

anemometer height [m], α is the power law exponent, $V_{(Z_{hub})}$ is the wind speed at the hub height of the wind turbine [m/s], $V_{(Z_{anem})}$ is the wind speed at anemometer height [m/s], and α , the power law exponent, which is a dimensionless parameter, is valued to 1/7.

- It refers to the wind turbine's power curve to calculate the power output under standard conditions of temperature and pressure. The power curve in Fig. 8 describes the amount of power the turbine produces versus the wind speed at hub height.
- It multiplies then the power output obtained from the wind turbine power curve by the air density ratio.

According to the ideal gas law, air density ratio can be expressed as follows (Adaramola et al., 2014, Hiendro et al., 2013):

$$\frac{\rho}{\rho_0} = \left(1 - \frac{BZ}{T_0} \right)^{\frac{gRB}{R}} \left(\frac{T_0}{T_0 - BZ} \right) \quad (6)$$

Where:

B is the lapse rate [0.00650 K/m], Z is the altitude [m], g is the gravitational acceleration [9.81 m/s²], T_0 is the standard temperature [288.16 K], R is a gas constant [287 J/kgK].

In other words, the actual air density of the site is expressed in the following equation (Mohammadi et al., 2014):

$$\rho = \rho_0 - (1.194 \times 10^{-4} Z_m) \quad (7)$$

Where Z_m is the site elevation in meters and ρ_0 is the air density under standard conditions (The air density at sea level at 1 atm and 60°F is 1.225 kg/m³).

In this study, the hub height of the wind turbine, Z_{hub} is 15 meters as the turbine may be located on the top of the roof of concrete urban house. The anemometer height was chosen at 10 meters above the ground. In this study, as the hub turbine is not the same as the anemometer height (10 meters), HOMER adjusts the wind speed data accordingly by using above equations to calculate the power output of the wind turbine.

6.3. Wind turbine and solar PV technical specification

To investigate whether a grid connected hybrid renewable system will satisfy the customer demand and save costs from using grid connected electricity, the authors are going to use the load data profile of household for the entire year, 8760, which is 43 kWh/day and a peak load of 9.2 kW. The cost data was collected from importer of solar PV system in local market of neighbouring country, Somalia (Somaliland), and the average cost the PV panel of 250Wc was found to be \$230. PV panels are made in Germany, called Copex solar, and are imported from Dubai. The initial cost of solar panels including all costs associated the PV (photovoltaic) systems such as modules, transport, mounting hardware, and installation, the replacement costs, and the annual O&M cost per 1 kW capacity including transport and installation were assumed to be \$1200, \$1000, and \$25, respectively.

The lifetime of PV panel was assumed to be 20 years. A de-rating factor of 80% is going to be used to calculate the electric generation from each PV panel leaving the 20% for dust and weather conditions such as temperatures effects. This study assumed that all panels were designed as fixed and tilted at slop degree of 11° 79' 130, which is equal to the latitude of Tadjourah city. A ground reflection was assumed to be 20%. The size of PV panels considered in this study were 0, 1, 2, 3, 4, 5, 6, 7, and 8 kW for the optimization analysis. General specification of both solar panels and wind turbine of the system is described in Tables 8 and 9, respectively.

The following Fig. 9 shows the structure of the proposed hybrid energy systems for the household load of 43kWh/day and its peak value of 9.2 kW. It describes the flow energy diagram and number of energy resources generators such PV, Wind and grid connection.

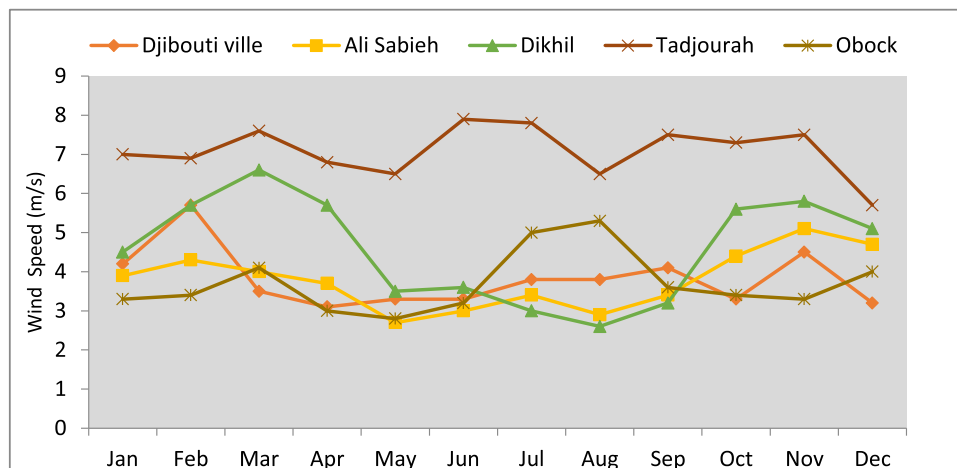


Fig. 7. Monthly mean wind speeds (m/s) for 5 Districts in Djibouti (m/s). Data from (Awaleh, 2007).

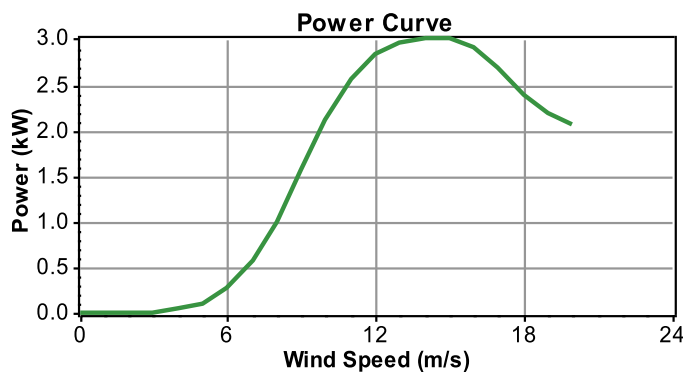


Fig. 8. Power curve of 3 kW wind turbine.

Table 8

Technical specification of photovoltaic panels from Copex solar.

COPEX Solar PV module	
Electrical Specification	Value and definition
Cell type:	Mono crystalline
Nominal maximum power:	250W
Number of solar module cells:	60 cells
Cell arrangement:	60 (6 * 10)
Optimum operating voltage:	30.7V
Optimum operating current:	8.14A
Open circuit voltage (Voc):	37.4V
Short circuit current (Isc):	8.96A
Power efficiency:	90% for the first 20 years
NOCT:	45°C
Mechanical specification	Value and definition
Operating temperature:	-40°C to +85°C
Temperature coefficient (Pmax):	-0.45/°C
Maximum system voltage:	1000V (IEC)/600V (UL)
Weights:	20kg
Dimensions per module:	1645 x 982 x 50mm
Total PV quantity needed:	32 * 250Wc Modules
Total area needed:	45m ²

6.4. Outputs' parameters for an economic analysis

In order to determine the optimal size of the components of this hybrid energy system for this study and to choose the best combinations of renewable generators to reduce consequently the cost of electric consumption from the grid and the total cost of the system, the authors bring into daylight some important parameters that HOMER software uses to get the required results such as the annualized cost of the system,

Table 9

General characteristics of considered Wind Turbine.

3 kW Wind Turbine			
N. of blade:	3	Efficiency:	80%
Blade length:	2.5 m	Max Speed:	600RPM
Rotor diameter:	5 m	Start Up Wind Speed:	2.5 m/s
Power:	3 kW	Wind Speed Cut:	25m/s
Tower height:	20 ft	Voltage:	96/240
Blades number:	3	Brand:	RRT Tech

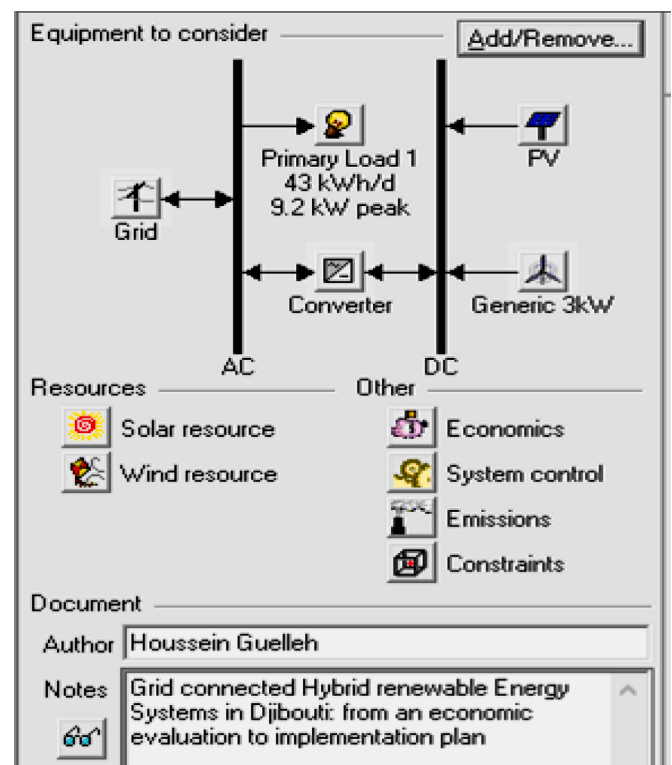


Fig. 9. The proposed grid connected hybrid renewable energy system.

the Net Present Cost (NPC), the Levelized Cost of Energy (COE) (or life-cycle cost), and the Capital Recovery Factor (CRF). HOMER uses total NPC as the primary economic figure in merit for optimization as it is a mathematical concept that does not require any arbitrary and disputable

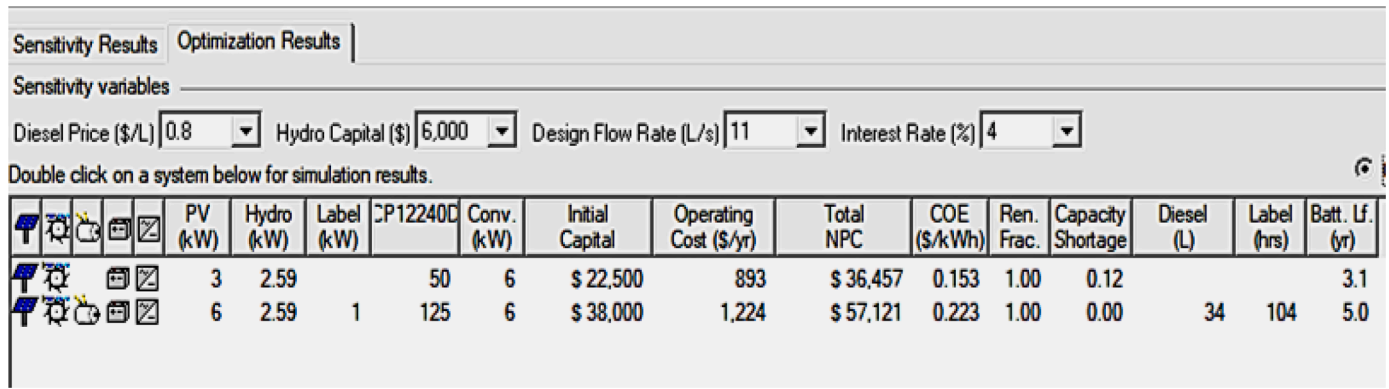


Fig. 10. Optimization results of hybrid energy system for a village in Cameroon.

judgements. The total NPC is HOMER's main economic output. HOMER ranks all systems according to total NPC.

The total annualized cost is an important value because HOMER uses it to calculate both the Levelized cost of Energy (LCOE) and the Net Present Cost (NPC). HOMER defines LCOE as the average cost per kWh of useful electrical energy produced by the system. However, HOMER shows as result in COE instead of LCOE. For the rest of our study, the authors are going to use only LCOE.

To calculate the LCOE, HOMER divides the annualized cost of producing electricity (the total annualized cost minus the cost of serving the thermal load) by the total useful electric energy production. It calculates the Levelized cost of Energy (LCOE) by the following equation (Çetinbaş et al., 2019):

$$LCOE = \frac{C_{ann, tot} - C_{boiler} E_{thermal}}{E_{prim, AC} + E_{prim, DC} + E_{def} + E_{grid, sales}} \quad (8)$$

Where:

$C_{ann, tot}$ is the total annualized cost of the system [\$/year], C_{boiler} is the boiler marginal cost [\$/kWh], $E_{thermal}$ is the total thermal load served [kWh/year], $E_{prim, AC}$ is the AC primary load served [kWh/year], $E_{prim, DC}$ is the DC primary load served [kWh/year], E_{def} is the deferrable load served [kWh/year], and $E_{grid, sales}$ is the total grid sale [kWh/year].

In this study, E_{def} is not applicable, and the system does not serve a thermal load ($E_{thermal} = 0$). Therefore, this term will equal to zero.

In other terms, Levelized Cost of Energy (LCOE) is expressed as follows:

$$LCOE = \frac{C_{tot}}{E_{tot}} \quad (9)$$

Where:

C_{tot} is the total annualized cost of the system [\$/Year] (cost of energy purchased from grid + initial cost + replacement cost + operation and replacement cost + fuel cost if applicable), and E_{tot} is the total Electricity consumption [kWh/year].

The capital recovery factor is a ratio used to calculate the present value of an annuity a series of equal annual cash flows). The equation for the capital recovery factors is expressed as follows (Adaramola et al., 2014, Masters, 2004):

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (10)$$

Where:

i is the real interest rate, and n is the number of years.

HOMER calculates the total net cost using the following equation:

$$C_{NPC} = \frac{C_{ann, tot}}{CRF(i, T_{proj})} \quad (11)$$

Where:

$C_{ann, tot}$ is the total annualized cost of the system [\$/yr], $CR(i, T_{proj})$ is the capital recovery factor, i is the annual real interest rate [%], and T_{proj} is the project lifetime.

7. Results and discussions

7.1. Results with HOMER validation

To investigate the results validity of this study with HOMER software, another study that had been done in 2009 Kenfack et al. (2009) has been selected randomly and studied for simulation and

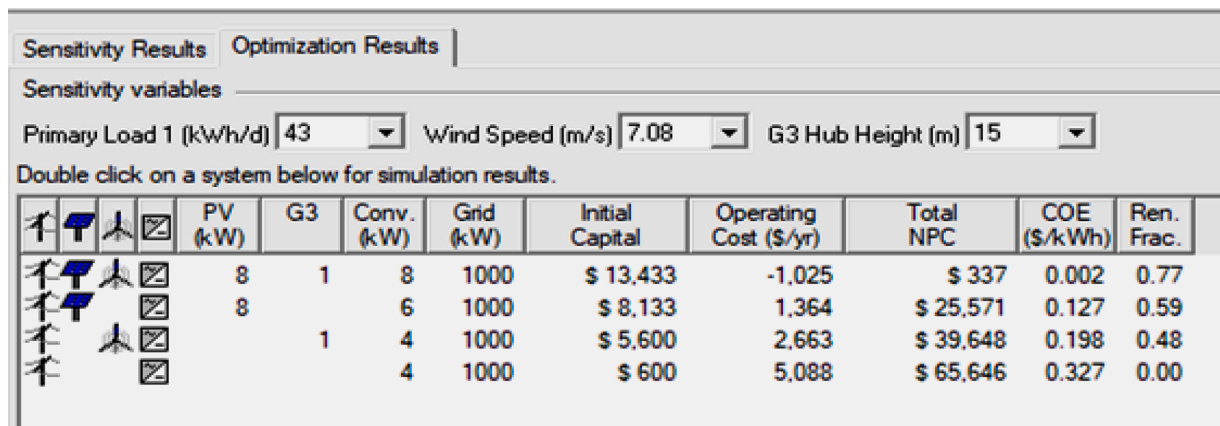


Fig. 11. Optimization results of hybrid system with 15 m hub height of WT.

Table 10
Hybrid system optimization results with different hub height of WT.

G3 Hub Height (m)	Operating cost (\$/year)	Total NPC (\$)	LCOE (\$/kWh)	Ren. Frac.
10	-768	\$3,619	0.018	0.76
15	-1,025	\$337	0.002	0.77
20	-1,197	\$-1,866	-0.009	0.78
25	-1,325	\$-3502	-0.017	0.79

optimization. The result obtained is almost similar with the result already obtained by Kenfack et al. and illustrated in Fig. 10. Therefore, the outcome of the present study with HOMER will be satisfactorily correct.

7.2. Results of optimization

7.2.1. Optimum grid connected hybrid system

In this section, the authors are going to give answer to the main objective of this study which was how to find an optimum grid connected hybrid energy system with the most suitable number of its renewable energy generators as components while minimising the cost of energy demand from the grid, and at the same time maximising the power reliability and availability for the proposed middle-class

household located in an urban area. Also, it should be kept in mind that implementing a hybrid renewable energy system in an urban area is totally different to its implementation in rural area where there is enough space for the installation. There will be some environmental constraints.

Thus, from the optimization results presented in Fig. 11, the optimum hybrid energy system that has been chosen for a middle-class household located in an urban area includes 8 kW of 250 Watt-peak of panels, one Wind Turbines (WT) of 3 kW each, 8 kW converter, and the grid connection. This optimum system has been wisely chosen due to the conditions that the system must meet to be feasible. In this study, due to limited space, the implementation of this study will be done on the top roof of concrete house. In other cases, the implementation of this hybrid energy system could be also done in houses with large gardens. However, the system can gain much space in high if there are no electrical cables crossing above the roof, i.e., the wind turbine can be installed on different hub heights to gain more energy. One should remember that the more the wind turbine is installed in higher hub the more it produces energy.

In this study, to see how the renewable energy penetration is affected and hence the total energy production, different hub heights of 10, 15, 20, 25 m were chosen as a sensitivity variable. According to the results shown in Table 10, the most suitable one would be the 15 m hub height

Table 11
NPC summary of grid connected hybrid system at 15m hub height WT.

Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
PV	7,233	2,556	2,556	0	-1,264	10,780
Generic 3kW	5000	624	2,556	0	-349	7,831
Grid	0	0	-21,160	0	0	-21,160
Converter	1,200	501	1,278	0	-93	2,886
System	13,433	3,380	-14,770	0	-1,707	337

Table 12
Output of Wind and PV grid connected hybrid system with 15 m height WT.

WT Quantity	value	Units	PV Quantity	Value	Units
Total rated capacity	3.00	kW	Rated capacity	8	kW
Mean output	1.05	kW	Mean output	1.61	kW
Capacity factor	35.0	%	Mean output	38.5	kWh/d
Minimum output	0.00	kW	Capacity factor	20.1	%
Maximum output	3.00	kW	Minimum output	0.00	kW
Wind penetration	58.5	%	Maximum output	7.77	kW
Hours of operation	7,690	hrs/yr	PV penetration	89.6	%
Total production cost	9,189	kWh/yr	Hours of operation	4,365	hrs/yr
Levelized cost	0.0667	\$/kWh	Total production	14,067	kWh/yr
			Levelized cost	0.0600	\$/kWh

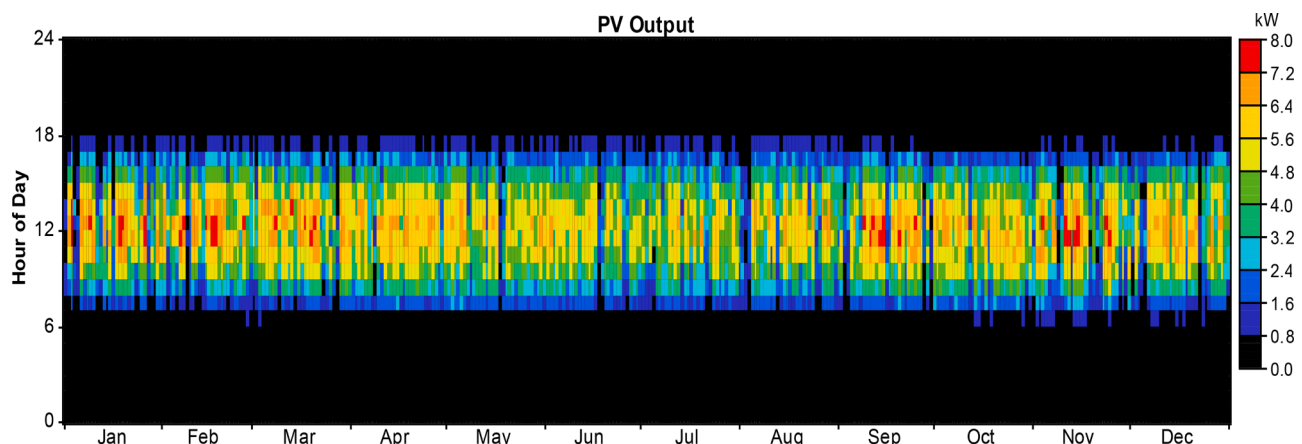


Fig. 12. PV output during the whole year.

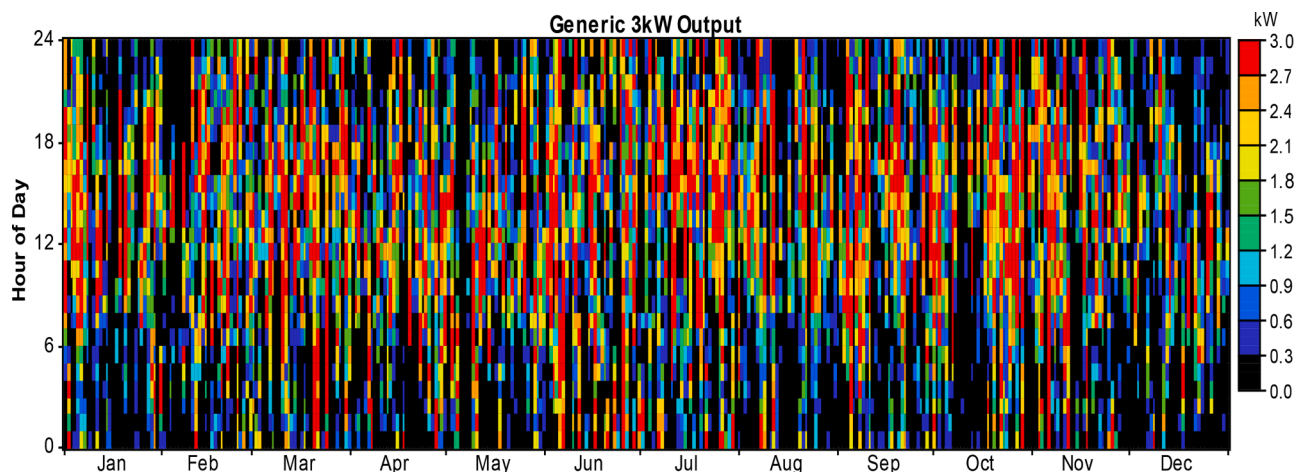


Fig. 13. Wind turbine output during the whole year.

Table 13
Converter output.

Quantity	Inverter	Units	Quantity	Inverter	Units
Capacity	8.00	kW	Hours of operation	7,999	hrs/year
Mean output	2.38	kW	Energy in	23,187	kWh/year
Minimum output	0.00	kW	Energy out	20,868	kWh/year
Maximum output	8.00	kW	Losses	2,319	kWh/year
Capacity factor	29.8	%			

for the WT in term of installation and suitability of urban landscape.

The initial capital cost of the components' system which is shown in Table 11 with the NPC summary comes to \$13,433. The total NPC for the most feasible configuration due to terrain limitation is \$337. The values for profit have negative signs such as the values for electricity sale to the power grid. The values for cost incurred are indicated by positive signs such as initial investment cost, replacement cost, etc.

As shown from the Table 12, the hours of operation of wind turbine, 7,690 hrs/year, are greater than the PV's ones, 4,365 hrs/year. This is mainly due to intermittency of solar and wind energy resources at Tadjourah during the whole year as can be shown in Figs. 12 and 13.

And finally, in terms of components output, a converter is required to convert AC load to DC acting as inverter or rectifier converting DC load form PV and WT to AC or both. In this study, it allows the DC power produced by the PV and Wind system to either supply on-site electrical household or to back-feed the grid when the PV/Wind system output is greater than the household load demand. As results show in Table 13, a converter of 8 kW capacity has been chosen for the current hybrid

system with total operation of 7,999 hours/year as illustrated in Fig. 14. Out of 23,187 kWh/year power production received from PV and Wind turbine, the converter has lost 2,319kWh/year during the process of converting and has successfully delivered 20,868kWh/year.

A data map of inverter or converter output, illustrated in Fig. 14 above, shows one year of hourly data of output energy which is coloured according to the data value of that hour during the year.

7.3. Discussions

From the optimization results presented in Fig. 11 above, it can be noticed that the main portion of electricity production is coming from the renewable energy generators, i.e., PV array and wind turbine, which represent a quarter of all energy produced. The electricity produced from Wind turbine when its mast is at 15 m height comes to 9,189 kWh/year. This represents 30% of the total production. One can bear in mind that the number of turbines were reduced to one of 3 kW maximum capacity due the environmental issues stated earlier such as space limitation, and urban landscape protection. And we should remember that the prospect site location has an excellent wind speeds all year around as shown in the following Fig. 15. In Fig. 15, the wind speeds go over 25 m/s over the course of the year, and the wind turbine's cut speed is 25 m/s to protect the turbine from damage. The wind is strong and available most of the time during the year. If there was enough space, with such wind speeds, the portion of electricity production from Wind turbine could be increased by increasing the number of turbines.

In results, the PV array represents 47% of this energy production. In total, the renewable energy fraction from this Wind-PV hybrid system comes to 77% as shown in Fig. 16 and Table 14. Only 23% of electricity is purchased from the grid.

The Table 14 summarises the total output of each energy producing

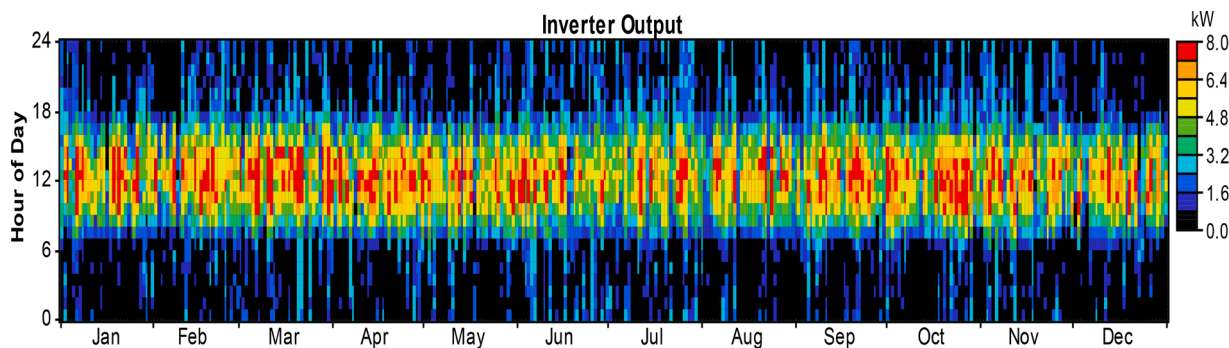


Fig. 14. DMap of Inverter/converter output.

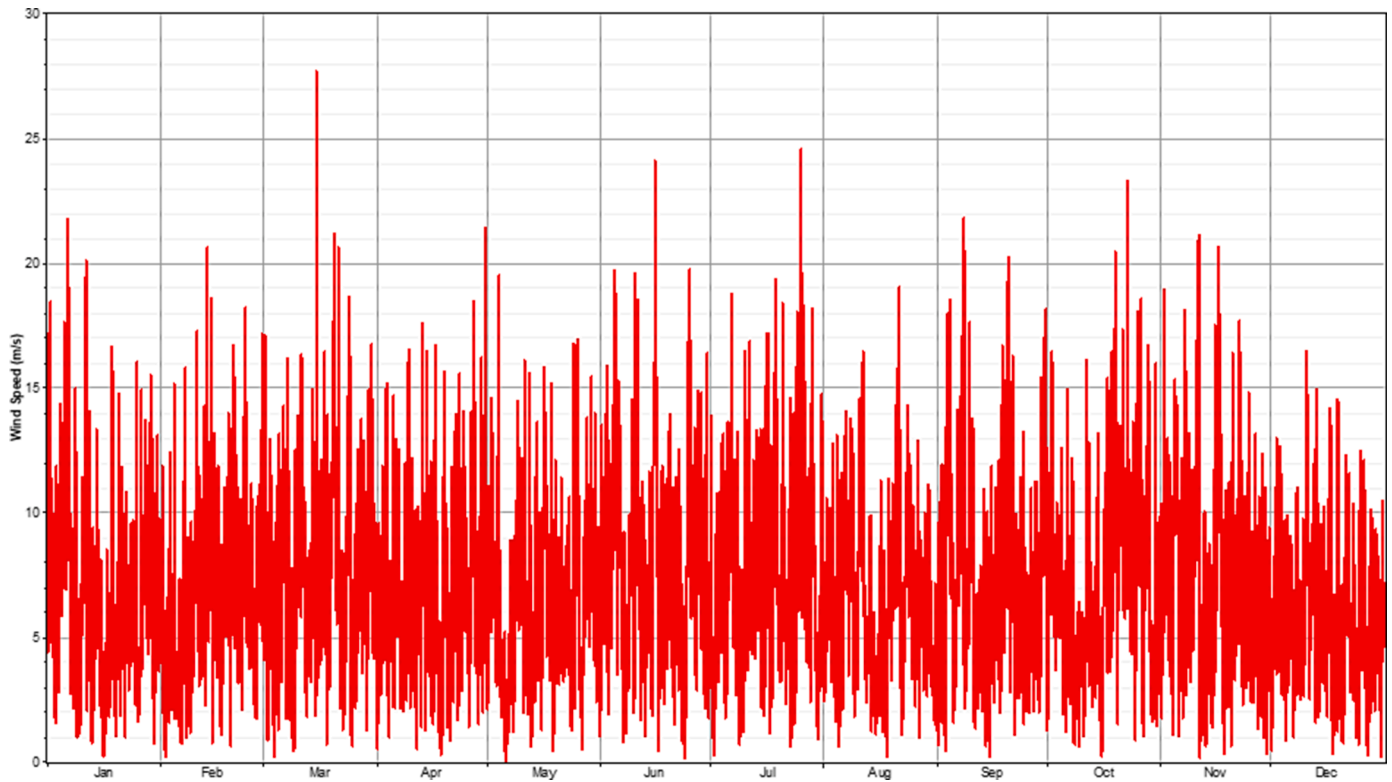


Fig. 15. Annual wind speeds of Tadjourah District.

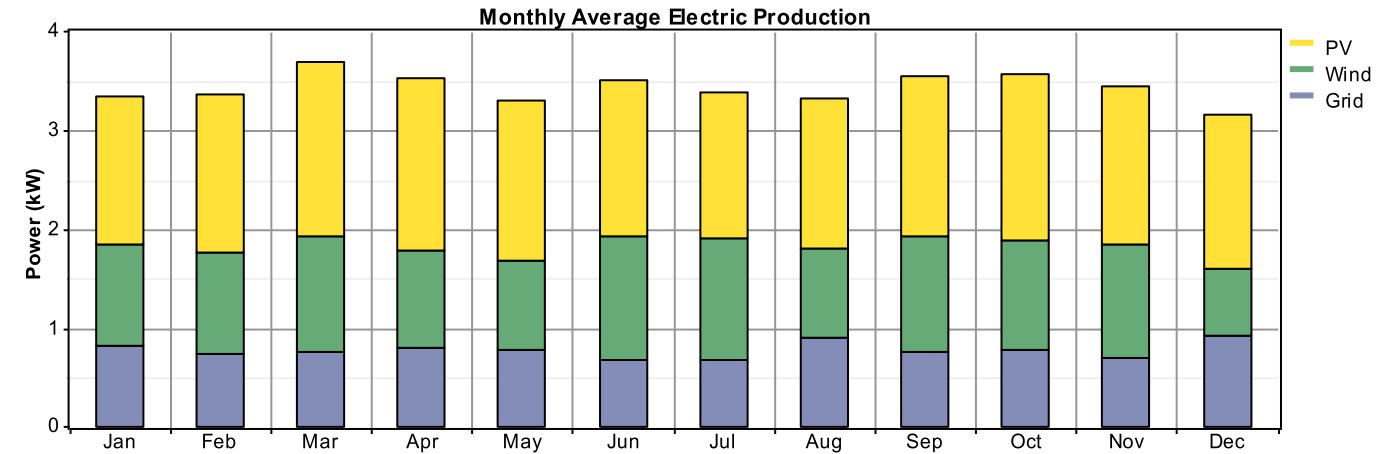


Fig. 16. Monthly average electrical production for the hybrid system.

Table 14
Optimal hybrid system's electricity components with WT at 15 m height.

Production	kWh/year	%	Consumption	kWh/year	%
PV array	14,067	47	AC primary load	15,695	57
Wind turbines	9,189	30	Grid sales	11,996	43
Grid purchases	6,822	23	Total	27,691	100
Total	30,078	100			

components of the optimal hybrid system including any annual electric loads served. In fact, the total production of electricity by all sources in the system comes to 30,078 kWh/year. The total amount of energy produced by PV array and wind Turbine with its mast 15 m height are 14,067 kWh/year and 9,189 kWh/year, respectively. The annual amount of energy that went serving the electrical load of the household,

the total amount of electricity sold to the grid during the year, and finally the total amount of electricity purchased from the grid come to 15,695 kWh/year, 11,996 kWh/year, and 6,822 kWh/year, respectively. Hence, the results of this study show that implementing a hybrid energy system in Tadjourah city is rewarding. The system firstly satisfies the household load and secondly sells electricity back to the grid which is even higher with the one purchased from it as illustrated in Fig. 17.

An optimal combination of renewable energy generators for the most economical schedule to satisfy the energy demand eliminates the power outage. The proposed hybrid renewable system can supply the household load with 24 h quality power without interruption.

The results show that the share of power generation by wind, PV and grid connection is a better solution than just using grid electricity. From the optimum simulation results, the total Net Present Cost (NPC), the Levelized Cost of Energy (LCOE) and the Operating cost of this system

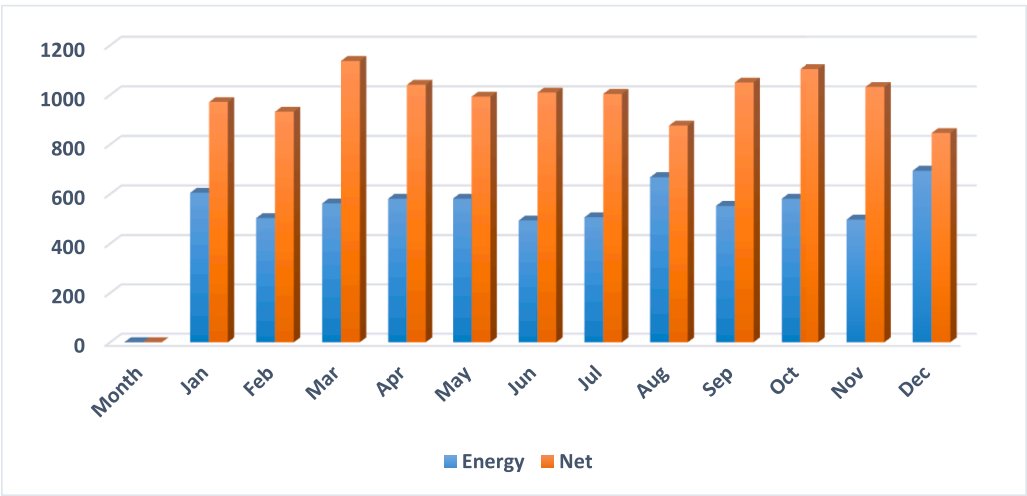


Fig. 17. Energy purchased from the grid Vs net energy sold to the grid.

Table 15
Hybrid system Vs grid for payback comparison.

	PV (kW)	G3 (kW)	Con. (kW)	Grid (kW)	Initial Capital	Total NPC
Base case (Grid)			4	1000	\$600	\$65,646
Current system	8	1	8	1000	\$13,433	\$337

Table 16
Hybrid system payback metric.

Metric	Value	Metric	Value
Present worth	\$ 65,310	Internal rate of return	48.5 %
Annual worth	\$ 5,109/year	Simple payback	2.06 years
Return on investment	47.7 %	Return on investment	2.27 years

when the mast of the wind turbine is 15 m height are \$337, \$0.002/kWh and \$-1,025/year, respectively. From these results, it can be concluded that the levelized cost of electricity generated from the hybrid system (\$0.002/kWh) is about 62.5% cheaper than the supplied electricity from the grid (\$0.32/kWh). Comparing both systems, the application of hybrid system has reduced 51% of the grid cost.

In this study, the initial investment is \$13,433. The grid connection is taken as a base case to know the number of the years that the new hybrid system brings back the value of the initial investment. The results are shown in Tables 15 and 16. From these results, the initial capital cost of \$13,433 will be paid back within 2.06 years with a return on investment of 47.7% of net electricity sold to the EdD. Fig. 18 validates visually these results showing that the system generates even income during the first year of its lifetime. The suggested optimal hybrid system was carefully chosen with the environmental constraints in urban landscape such as roof top installation with NPC of \$337, LCOE of \$-0.002/kWh, and operating cost of \$-1,025/year. Compared with grid-only connection of \$0.32/kWh, the optimal hybrid system chosen for this study is

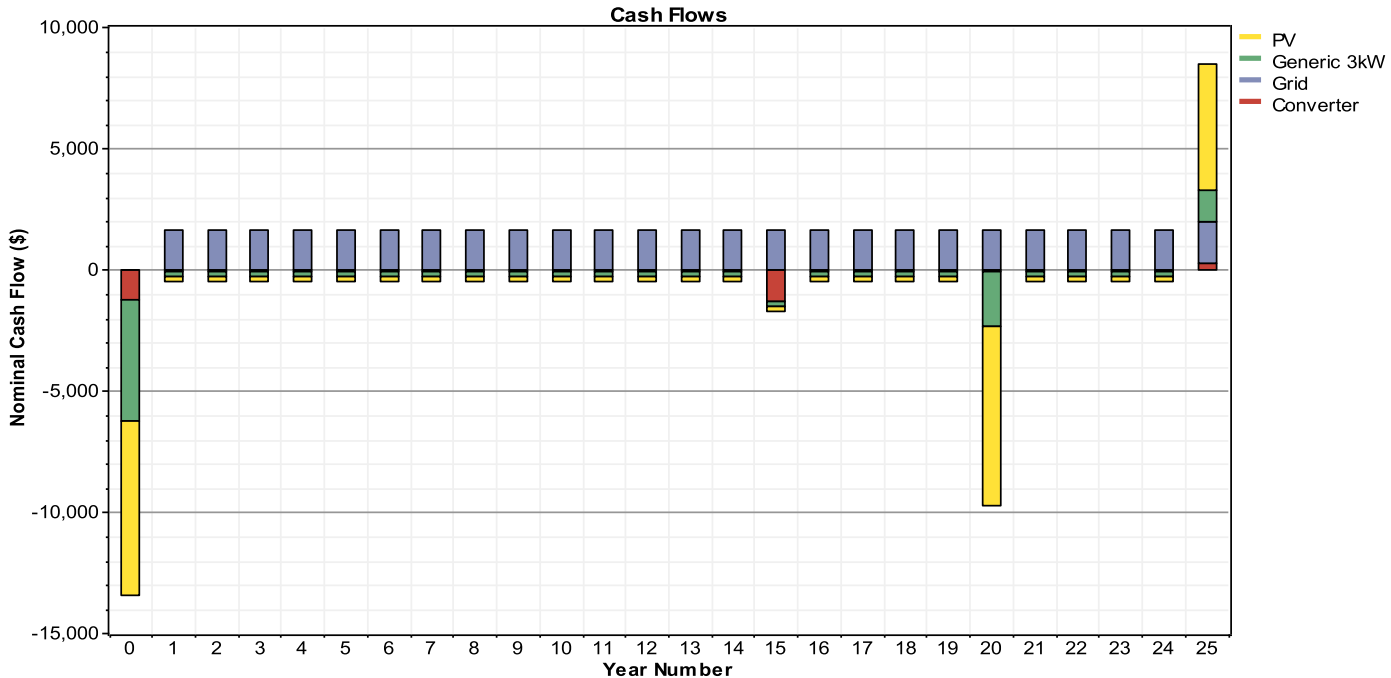


Fig. 18. Cash flow by component of the hybrid system.

Table 17
Emissions outputs.

Pollutant	Emissions (kg/year)	Pollutant	Emissions (kg/year)
Carbon dioxide	-3,269	Particulate matter	0
Carbon monoxide	0	Sulfur dioxide	-14.2
Unburned hydrocarbons	0	Nitrogen oxides	-6.93

going to reduce 51% of the grid cost when it is implemented.

In this present study, both wind turbine and PV array in Fig. 16 have high capacity of power production. Tadjourah city receives more solar irradiation in March, April, and May (Fig. 5) and a good potential and higher speed wind through the whole year specially in June and July (Fig. 6). This could neutralize the air conditioning cooling load high in this period

7.3.1. Emissions output

During its lifetime, the hybrid energy system connected to the grid produces emissions because the electricity obtained from the grid has been produced by thermal generators consuming fossil fuels. Table 17 shows the total amount of each pollutant produced annually by the power system. Selling power to the grid reduces grid emissions. If the system sells more power to the grid than it buys from the grid over the year, the net grid purchases will be negative and so will the grid-related emissions of each pollutant. To calculate the emissions of each pollutant associated with these net grid purchases, the net grid purchases (in kWh) is multiplied by the emission factor 9 g/kWh for each pollutant. Thus, the optimal hybrid system with hub height 15 m decreases car dioxide emissions by over 3.2 tonnes per year.

8. Conclusion and recommendations

In the present study, the main objective was to find possible way to reduce or neutralise the burden of huge cost of grid supplied electricity bill for an urban household in Tadjourah city located in the North-Eastern of Djibouti. For this purpose, an economic evaluation of different hybrid grid-connected energy systems (HRES) has been carried

out using HOMER Software. The optimal HRES chosen for this study is Wind-PV grid connected system. The initial investment cost of this hybrid energy system is \$13,433, a very low investment comparing to other studies (Alharthi et al., 2018), (Vakilabadi et al., 2019), (Abdelshafy et al. 2017), and (Vuc et al., 2011), with renewable energy generators containing 8 kW of photovoltaic modules, one Wind turbine of 3 kW with hub height of 15 m, and 8 kW of Converter. This optimal HRES represents a proportion of 77% of renewable energy contribution (47% and 30% of Solar and wind penetration, respectively) and power grid of 23%. Compared with grid-only connection of \$0.32/kWh, the optimal hybrid system chosen for this study is going to reduce 51% of the grid cost when it is implemented.

The present study seeks attention firstly of customers who live in urban areas in developing countries, and secondly stakeholders, investors, and policy makers. The proposed methodology is general and can be applied in any urban household in developing countries where the electricity is unreliable, unavailable, and very expensive. However, there is a limitation about this methodology. Due to the environmental landscape and short space, it is recommended that the number of the turbines to be installed on the rooftop of a concrete urban house should be limited to few if not one. In the context of Djibouti, it is recommended also that the sale of electricity excess to EdD should be checked with the energy authority as the implementation policy is still being processed up to present-day by the Ministry of Energy of Djibouti.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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APPENDIX A: Repartition of hours of each appliance for Daily electrical load of a typical F4 middle class household in Djibouti

Living room	Fluorescent tubes (x2) 1x40Wx5 Hours+2x40x7 Hours	Television (x1) 1x120Wx 11 Hours	Ceiling fans (x2) 1x 80Wx3 Hours +2x80x7 Hours	Split air conditioner (x1) 1x2400x3 Hours	Desktop PC (x1) 1x90x2.5hours
Bedroom	Fluorescent tube (x1) 1 x40Wx4 Hours	Split air conditioner x1 1x2400x8 Hours	Ceiling fan (x1) 1x 80Wx10 Hours		
2nd Room	Fluorescent tube (x1) 1x40Wx4 Hours	Ceiling fan (x1) 1x 80Wx 11 Hours			
3rd Room	Fluorescent tube (x1) 1x40Wx3 Hours	Ceiling fan (x1) 1x80Wx11 Hours			
Kitchen	Fluorescent tube (x1) 1x40Wx5 Hours	Blinder Moulinex (x1) 1x500Wx0.25 Hours	Microwave (x1) 1x700x0.25 Hours	Fridge (x1) 1x250Wx24 Hours	
Other appliances and gadgets	Fluorescent Tube- Front, Back and corridor (x4) 4x40Wx7 Hours	Mobiles (x4) and tablets (x3) 4x18+18x3 Hours	Ironing (x1) 1x1100x0.75 Hours	Washing machine (x1) (x1)1x500x2.5 Hours	
Toilet	Fluorescent tube (x1) 1x40x3 Hours				

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